

🚗 Adult Income Dataset (1994)

1 DAX Assignment

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Link to DASHBOARD:

- <https://app.powerbi.com/view?r=eyJrIjoiMTg3YmRhMGItNDdlMS00NjUxLTk3MDctMWRmZmMxYWVhM2E5IiwidCI6IjE2ZDgzZWU2LTI1NGEtNDY5ZC1hNmNjLTU0ZTJjYTYiMTNlNyIsImMiOiJh9>

Source: UC Irvine Machine Learning repository **Dataset:** Adult **Provider:** 1994 census dataset **Rows:** 48842

Dataset link: <https://archive.ics.uci.edu/dataset/2/adult>

2 Proof of Website

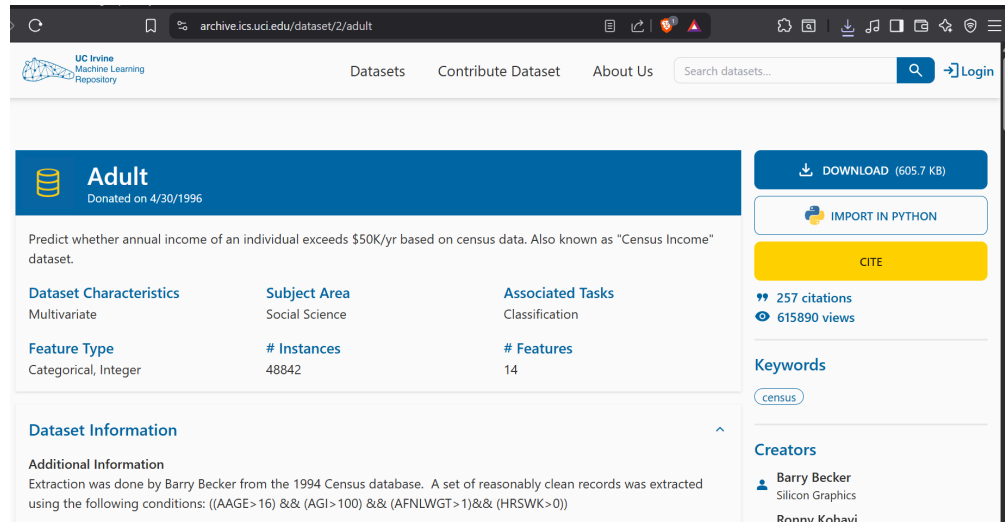


Fig. 1 proof of Data/website

3 Dataset

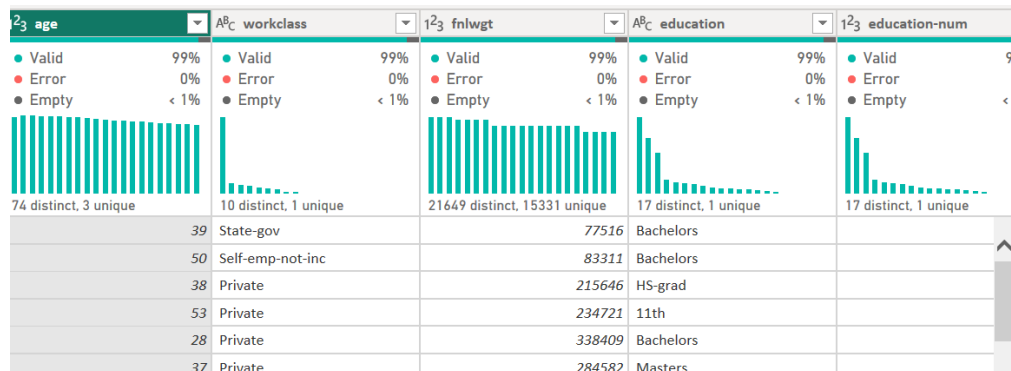
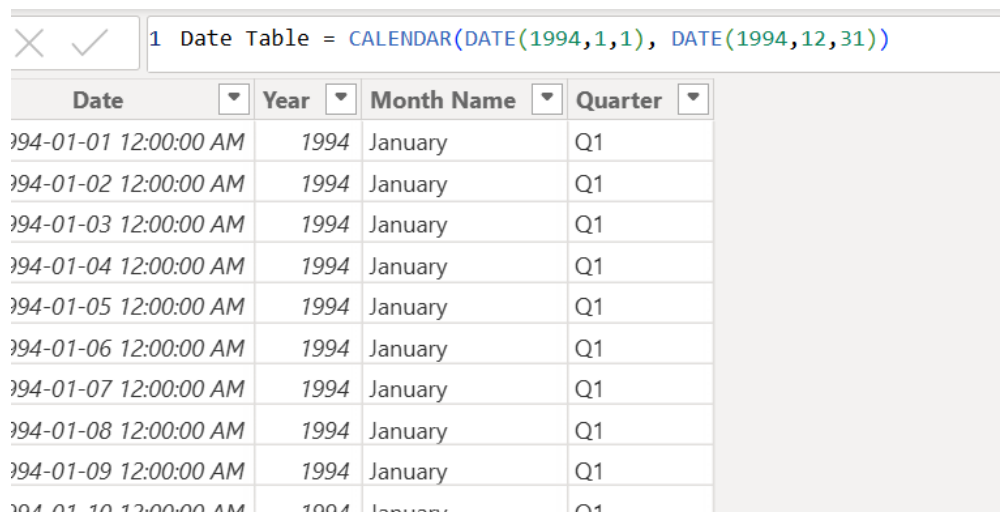


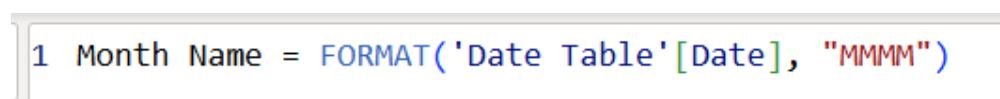
Fig. 2 dataset_example

4 Creating Date Table



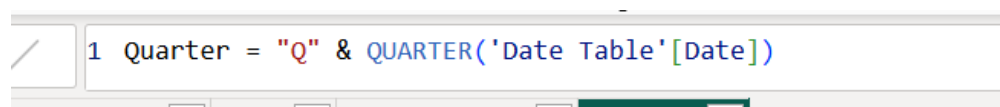
Date	Year	Month Name	Quarter
1994-01-01 12:00:00 AM	1994	January	Q1
1994-01-02 12:00:00 AM	1994	January	Q1
1994-01-03 12:00:00 AM	1994	January	Q1
1994-01-04 12:00:00 AM	1994	January	Q1
1994-01-05 12:00:00 AM	1994	January	Q1
1994-01-06 12:00:00 AM	1994	January	Q1
1994-01-07 12:00:00 AM	1994	January	Q1
1994-01-08 12:00:00 AM	1994	January	Q1
1994-01-09 12:00:00 AM	1994	January	Q1
1994-01-10 12:00:00 AM	1994	January	Q1

Fig. 3 created date table dax



1 Month Name = FORMAT('Date Table'[Date], "MMMM")

Fig. 4 monthdax



1 Quarter = "Q" & QUARTER('Date Table'[Date])

Fig. 5 quarterdax

5 Creating relationship

From table
Date Table

Date	Month Name	Quarter	Year
01/01/1994 1...	January	Q1	1994
02/01/1994 1...	January	Q1	1994
03/01/1994 1...	January	Q1	1994

To table
adult

age	capital-gain	capital-loss	Date	education	education-num	fnlwgt
49	0	0	10/09/1994 1...	HS-grad	9	193366
36	0	0	04/05/1994 1...	HS-grad	9	155537
28	0	0	26/06/1994 1...	HS-grad	9	92262

Cardinality
One to many (1:*)

Cross-filter direction
Single

☒ Make this relationship active
☐ Assume referential integrity

☐ Apply security filter in both directions

Save Cancel

Fig. 6 creating relationship between date table and main fact table

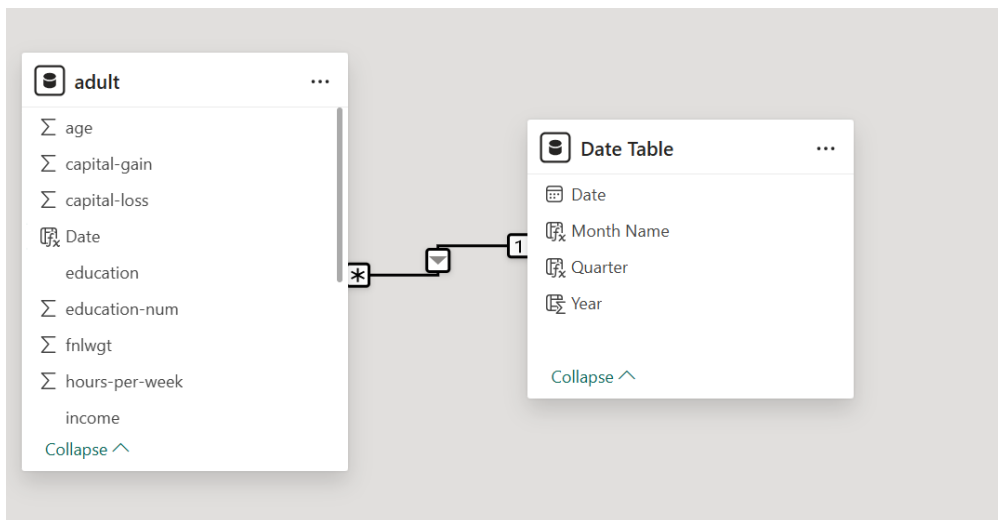


Fig. 7 star schema

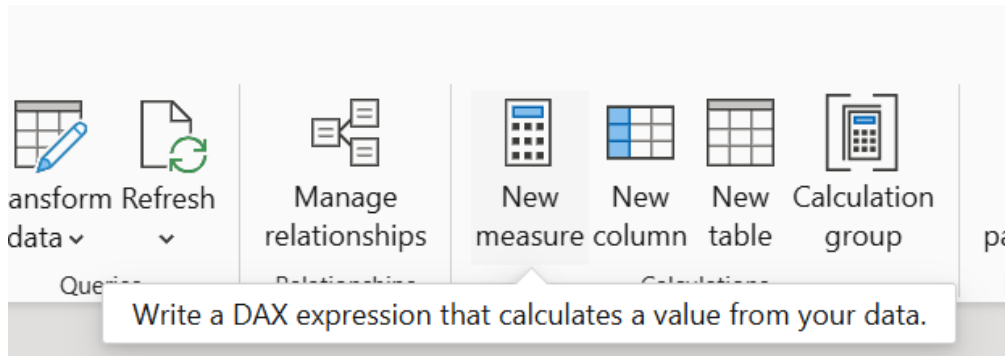


Fig. 8 where the new measures is

- 6 Measure 1: Total individuals

```
1 Total Individuals = COUNTROWS('adult')
```

Fig. 9 total individuals dax

- 7 Measure 2: Average worked hours

```
1 Avg Hours per Week = AVERAGE('adult'[hours-per-week])
```

Fig. 10 average worked hoursdax

- 8 Measure 3: High Income count

```
1 High Income Count = CALCULATE([Total Individuals], 'adult'[income] = ">50K")
```

Fig. 11 high income count dax

- 9 Measure 4: High income percentage

```
1 High Income % = DIVIDE([High Income Count], [Total Individuals], 0)
```

Fig. 12 high income % dax

- 10 Measure 5: Low income Count

```
1 Low Income Count = CALCULATE([Total Individuals], 'adult'[income] = "<50K")
```

Fig. 13 low income count dax

- 11 Column 1: Age Group Classification

```
Age Group =  
SWITCH(TRUE(),  
    'adult'[age] < 25, "Young (18-24)",  
    'adult'[age] <= 34, "Early Career (25-34)",  
    'adult'[age] <= 44, "Mid Career (35-44)",  
    'adult'[age] <= 54, "Senior (45-54)",  
    'adult'[age] <= 64, "Pre-Retirement (55-64)",  
    "Retirement Age (65+)")
```

Fig. 14 age group dax

12 Column 2: Work Hours Category

```
1 Work Hours Category =  
2 SWITCH(TRUE(),  
3     'adult'[hours-per-week] < 20, "Part Time",  
4     'adult'[hours-per-week] <= 40, "Full Time",  
5     'adult'[hours-per-week] <= 50, "Overtime",  
6     "Extreme Hours"  
7 )
```

Fig. 15 work hours category

13 Column 3: Education Level Group

```
1 Education Group =  
2 SWITCH('adult'[education],  
3     "Preschool", "No High School",  
4     "1st-4th", "No High School",  
5     "5th-6th", "No High School",  
6     "7th-8th", "No High School",  
7     "9th", "No High School",  
8     "10th", "No High School",  
9     "11th", "No High School",  
0     "12th", "High School Graduate",  
1     "HS-grad", "High School Graduate",  
2     "Some-college", "Some College",  
3     "Assoc-acdm", "Associate Degree",  
4     "Assoc-voc", "Associate Degree",  
5     "Bachelors", "Bachelor's Degree",  
6     "Masters", "Master's Degree",  
7     "Prof-school", "Professional Degree",  
8     "Doctorate", "Doctorate Degree"  
9 )
```

Fig. 16 edu levl dax

14 Measure 1: Female High Income percentage

```
1 Female High Income % =  
2 VAR FemaleHighIncome = CALCULATE([High Income Count], 'adult'[sex] = "Female")  
3 VAR TotalFemale = CALCULATE([Total Individuals], 'adult'[sex] = "Female")  
4 RETURN DIVIDE(FemaleHighIncome, TotalFemale, 0)
```

Fig. 17 female high income percentage

15 Measure 2: Percentage of National Average

```
Income vs National Avg =
VAR CurrentAvgHours = [Avg Hours per Week]
VAR NationalAvgHours = CALCULATE([Avg Hours per Week], ALL('adult'))
RETURN
DIVIDE([CurrentAvgHours - NationalAvgHours], NationalAvgHours, 0)
```

Fig. 18 % of national income

16 Measure 3: Professional Contribution

```
1 Professional % of Total =
2 VAR ProfCount = CALCULATE([Total Individuals], 'adult'[occupation] = "Prof-specialty")
3 RETURN DIVIDE([ProfCount], [Total Individuals], 0)
```

Fig. 19 professional contribution

17 Measure 4: Gender Gap Analysis

```
. Gender Income Gap =
. [High Income %] - [Female High Income %]
```

Fig. 20 gender income gap dax

18 Total Estimated Income (SUMX)

```
1 Total Estimated Income =
2 SUMX(
3     'adult',
4     'adult'[hours-per-week] * 20 * 52
5 )
```

Fig. 21 SUMX dax

19 Average Education Level by Occupation (AVERAGEX)

```
. Avg Education by Occupation =
. AVERAGEX(
.     VALUES('adult'[occupation]),
.     AVERAGE('adult'[education-num])
. )
```

Fig. 22 averagexdax

20 Total Capital Activity (SUMX with multiple columns)

```
1 Total Capital Activity =  
2 SUMX(  
3     'adult',  
4     'adult'[capital-gain] + 'adult'[capital-loss]  
5 )
```

Fig. 23 sumxMultiDAX

21 Measure 1: Year-to-Date Individuals

```
1 YTD Individuals =  
2 TOTALYTD([Total Individuals], 'Date Table'[Date])
```

Fig. 24 time intel dax 1

22 Measure 2: Previous Quarter (comparing to last period)

```
1 Previous Quarters Individuals =  
2 CALCULATE([Total Individuals], PREVIOUSQUARTER('Date Table'[Date]))
```

Fig. 25 time intel dax 2

23 Measure 3: Quarter-over-Quarter Growth percentage

```
QoQ Growth % =  
VAR CurrentQ = [Total Individuals]  
VAR PreviousQ = [Previous Quarter Individuals]  
RETURN DIVIDE(CurrentQ - PreviousQ, PreviousQ, 0)
```

Fig. 26 time intel dax 3

24 Measure 4: Same Period Last Year

```
SPLY Individuals =  
CALCULATE([Total Individuals], SAMEPERIODLASTYEAR('Date Table'[Date]))
```

Fig. 27 time intel dax 4

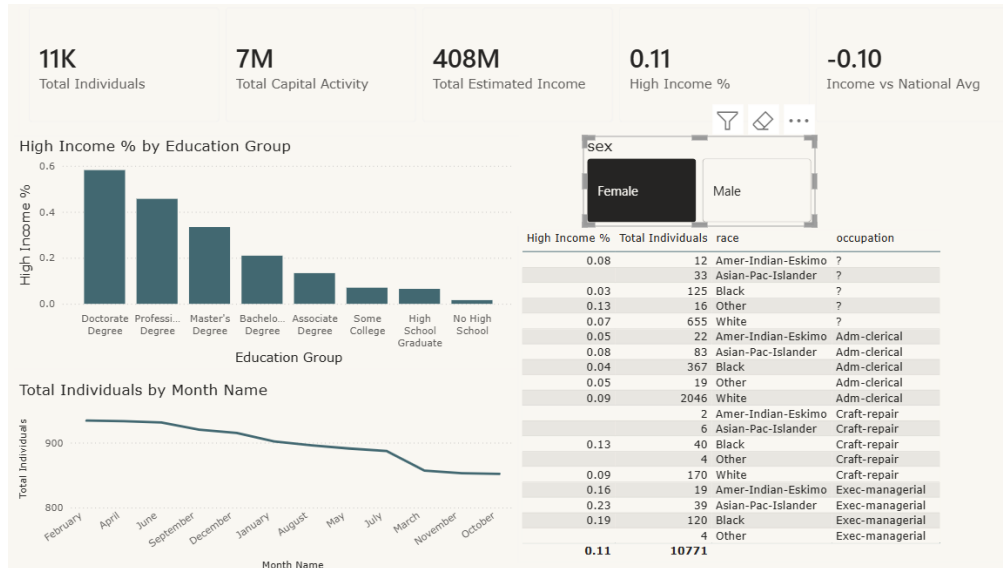


Fig. 28 visual

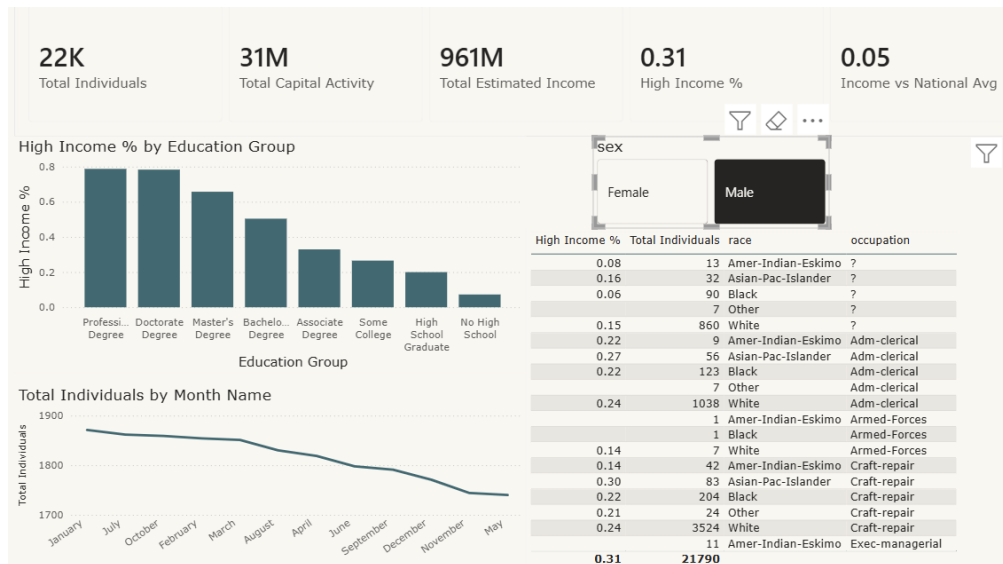


Fig. 29 alt text